A Controller Comparison

Topic 7.7 · TODAY'S PROBLEM

Suppose a cooperative has 1,200 cases with conventional controllers and 900 with adaptive controllers. It takes independent SRSs of 40 existing cases from each group; controller type was not assigned. A 95% interval for $\mu_C - \mu_A$ is $(1.219, 4.781)$ kWh per case-day. A retrofit is worthwhile only if mean savings are at least 2.0 kWh.

Rhea: "Zero is excluded and 2 is inside, so retrofitting is shown to cause savings of at least 2 kWh."

Sol: "I would compare the interval separately with 0 and 2, then inspect sampling and assignment before applying it to a retrofit."

Daily kWh samples

Controller	N	n	$\bar{x}$	s
Conv.	1,200	40	18.6	4.0
Adaptive	900	40	15.6	4.0

FIRST TAKE (5 min, on your sheet)

Which account is better supported so far? Identify any interval feature or design detail needed to decide.

DOK 1 **(a)** Define $\mu_C - \mu_A$, calculate the point estimate, and locate 0 and 2 relative to the interval.

DOK 2 **(b)** Verify independent sampling, both 10 percent checks, and large-sample shape. Reproduce SE, df, margin of error, and the interval; interpret it.

DOK 3 **(c)** Adjudicate the accounts using the positions of 0 and the 2-kWh threshold. Bound population and causal scope from sampling and assignment, and explain the budgeting consequence of the rejected interpretation.

Video + follow-along: `u7_lesson7_live.html` (scan the code)
Finish (a)–(c) after the video. Turn in your sheet.